

Fall 23, 4100 Problem Set 9

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Find the spectral decomposition of the self-adjoint matrix

$$A = \begin{bmatrix} 4 & 1+i \\ 1-i & 3 \end{bmatrix}$$

2. Let A be hermitian and let λ be a (real) eigenvalue of A . Prove that $\text{null}((A - \lambda I)^2) = \text{null}(A - \lambda I)$.
3. Let A be a symmetric, positive definite, $n \times n$ matrix and let B be an $n \times m$ matrix. Prove that $B^T A B$ is positive definite if and only if $\text{rank}(B) = m$.
4. Use linear algebra to find the minimum value of the paraboloid, $f(x_1, x_2) = 2x_1^2 + 4x_1x_2 + 3x_2^2 - 4x_1 + 5x_2 + 8$. Be sure to justify why the value you obtain is indeed a minimum.
5. Consider the Rayleigh quotient, $R(x_1, x_2) = \frac{5x_1^2 + 6x_1x_2 + 13x_2^2}{x_1^2 + x_2^2}$ where $\mathbf{x} \neq \mathbf{0}$. Find a symmetric matrix A such that $R = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$ and then find the minimum and maximum values of R . Be sure justify why these values are indeed the extreme values of R .
6. Let A and B be self-adjoint, positive definite matrices. Prove that the eigenvalues of AB are also real and positive.